

Medical RAG AI for Ophthalmology Follow-up: Implementation and Evaluation

1. Introduction

This report presents the development and evaluation of a Retrieval-Augmented Generation (RAG) system designed for medical follow-up assistance in ophthalmology and refractive surgery. The system addresses a critical challenge in healthcare: providing patients with accurate, evidence-based answers to common post-operative questions while maintaining medical rigor and reducing physician workload.

The core motivation stems from the repetitive nature of patient inquiries following refractive surgery procedures. Physicians frequently answer the same questions about recovery timelines, potential complications, and post-operative care. A RAG-based system can provide consistent, reference-backed responses while freeing medical professionals to focus on complex cases requiring human judgment.

This report is structured as follows: Section 2 describes the medical document dataset and preprocessing pipeline. Section 3 details the NLP architecture, including embedding models and retrieval mechanisms. Section 4 outlines the experimental setup for comparing RAG-enhanced responses against baseline LLM outputs. Section 5 discusses hyperparameter optimization and model validation strategies. Section 6 presents evaluation results based on physician scoring of system responses. Section 7 summarizes findings and proposes future improvements.

2. Dataset Description

2.1 Data Sources

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The dataset consists of 800 medical documents totaling 2,265,194 characters, comprising 784 PDF files from ophthalmology textbooks and clinical practice guidelines, and 16 Microsoft Word documents containing supplementary clinical protocols. Primary sources include the American Academy of Ophthalmology's Preferred Practice Patterns (PPP) for refractive surgery, peer-reviewed clinical studies, and standardized patient care protocols. Documents cover topics including LASIK, PRK, SMILE procedures, pre-operative screening criteria, intra-operative protocols, and post-operative management.

2.2 Document Preprocessing

Large PDF files (some exceeding 500MB) were split into manageable chunks using a threshold-based strategy. Files over 30 pages were segmented into 10-page chunks with 1-page overlap, producing a final dataset where all files are under 7MB. The original 37 large files expanded to 800 processable documents.

2.3 Multi-modal Content Extraction

Beyond text extraction, the preprocessing pipeline handles tables and images. Tables were extracted using PyMuPDF's table detection, converted to markdown format, and embedded as text with metadata indicating tabular structure. Medical diagrams and anatomical illustrations were extracted with bounding box coordinates, OCR was applied to capture embedded text, and image descriptions were stored as metadata alongside text chunks. This multi-modal approach ensures that critical information presented in non-text formats (such as surgical outcome tables or anatomical diagrams) is incorporated into the retrieval system.

2.4 Chunking Strategy

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Documents were segmented using recursive character splitting with a chunk size of 800 characters and 100-character overlap, generating 4,119 total chunks (averaging 45 chunks per document). This configuration balances context preservation with retrieval precision. Smaller chunks risk fragmenting medical concepts, while larger chunks dilute similarity matching. The 100-character overlap ensures continuity across semantic boundaries. Each chunk retains metadata including source document, page number, content type (text/table/image-caption), and document section hierarchy, enabling provenance tracking and source citation in responses.

3. NLP Model and Algorithm

3.1 System Architecture

The RAG system employs a two-stage architecture: retrieval and generation. Given a patient query, the system first identifies relevant document chunks through semantic similarity search, then generates a response by conditioning a language model on both the query and retrieved context. Formally, for query q , the system retrieves top- k documents $D = \{d_1, d_2, \dots, d_k\}$ and generates response r :

$$D = \operatorname{argmax}_{\{d \in \text{Corpus}\}} \text{similarity}(\text{embed}(q), \text{embed}(d))$$

$$r = \text{LLM}(\text{prompt}(\text{system_role}, D, q))$$

3.2 Embedding Model

Document and query embeddings are generated using BAAI/bge-base-zh-v1.5, a 768-dimensional dense retrieval model optimized for Chinese text but effective for English medical terminology. The model architecture is based on BERT with 12 transformer layers, 768 hidden dimensions, 12 attention heads, and a maximum sequence length of 512 tokens. Embedding

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generation employs mean pooling over token representations: $e = \text{mean}([h_1, h_2, \dots, h_n])$, where h_i represents the hidden state of token i from the final transformer layer.

The model was selected for its strong performance on cross-lingual medical terminology, as ophthalmology literature contains extensive Latin-derived technical terms common across languages. GPU acceleration (CUDA) reduced embedding computation time from approximately 1 second per query on CPU to 0.011 seconds on A10G GPU, representing a 100-fold speedup critical for real-time applications.

3.3 Vector Database

ChromaDB serves as the vector store, implementing approximate nearest neighbor search through HNSW (Hierarchical Navigable Small World) indexing. For a query embedding q , similarity is computed using cosine distance: $\text{similarity}(q, d) = (q \cdot d) / (\|q\| \times \|d\|)$. The retrieval function returns documents where similarity exceeds a threshold $\tau = 0.5$, ranked by descending similarity score. The top- k parameter ($k=2$ for most experiments) controls the number of chunks passed to the language model.

3.4 Language Model

The system uses DeepSeek-R1:1.5b (local deployment via Ollama) with 1.5 billion parameters and a 934MB model size optimized for inference speed. The system employs role-based prompting with a medical persona emphasizing compassion, evidence-based information, clear patient-appropriate language, and clinical rigor. The prompt structure for RAG-enhanced responses includes the system role, medical references from the knowledge base with metadata, the patient question, and instructions to provide a professional medical response based on the

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retrieved references. Response length is controlled to 150-200 words to balance completeness with patient comprehension.

3.5 Algorithm Development Background

The RAG paradigm addresses knowledge limitations in parametric language models by decoupling knowledge storage (vector database) from reasoning (language model). This enables dynamic knowledge updates without retraining, explicit source attribution, and reduced hallucination through grounding in retrieved documents. The medical domain particularly benefits from RAG due to rapidly evolving clinical guidelines requiring frequent knowledge updates, legal and ethical requirements for traceable evidence-based responses, and specialized terminology exceeding general-purpose LLM training data.

4. Experimental Setup

4.1 Evaluation Framework

The experiment compared two conditions across 25 common patient questions about refractive surgery: (1) Without RAG - LLM generates responses using only parametric knowledge from model weights, and (2) With RAG - LLM generates responses conditioned on retrieved medical documents. Both conditions use identical system prompts to ensure fair comparison, with the key difference being the presence or absence of the "Medical References from Knowledge Base" section in the prompt.

4.2 Test Questions

The 25 evaluation questions were provided by a practicing ophthalmologist based on frequently asked patient questions. Questions span general procedure information, eligibility criteria, safety

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and complications, recovery and outcomes, and procedure comparisons, representing realistic clinical scenarios encountered in post-operative care.

4.3 Implementation

The system was deployed on AWS EC2 with A10G GPU (NVIDIA A10G with 24GB VRAM, 16 vCPUs, 64GB RAM, Ubuntu 24.04, CUDA 12.1). Key implementation components include LangChain for RAG orchestration, ChromaDB with persistent storage, HuggingFace Transformers with CUDA acceleration, and Ollama serving DeepSeek-R1:1.5b locally.

4.4 Performance Metrics

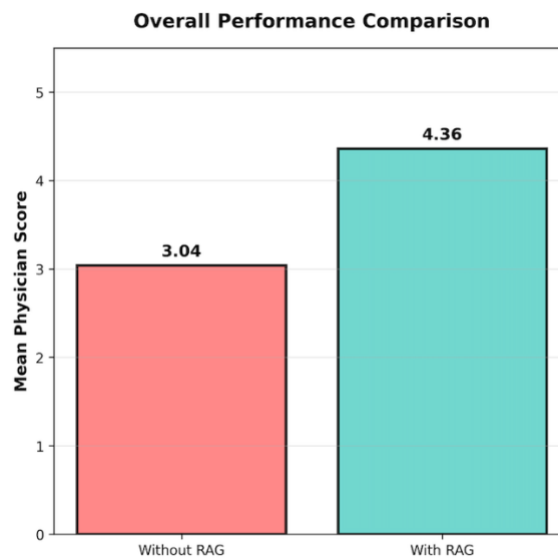


Figure 1

System performance was evaluated along multiple dimensions. Quantitative metrics included response time (end-to-end latency), prompt length (character count of input to LLM), and answer length (character count of generated response). Qualitative metrics included medical accuracy (physician rating on 5-point scale: 1=incorrect/dangerous, 5=accurate and complete), clinical

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appropriateness (evaluation of language, empathy, and actionability), and source attribution (presence and quality of cited references in RAG condition only).

Each question was evaluated by a licensed ophthalmologist who assigned scores based on factual correctness, completeness of information, appropriateness for patient understanding, and absence of potentially harmful advice. The physician was blinded to condition (With/Without RAG) during scoring (FIGURE 1).

4.5 Baseline Comparison Rationale

The "Without RAG" baseline represents current common practice: patients searching general-purpose LLMs for medical information. By comparing against this baseline, we measure the added value of domain-specific knowledge retrieval versus relying on LLM parametric knowledge alone.

5. Hyperparameter Optimization and Validation

5.1 Retrieval Hyperparameters

The number of retrieved documents (top-k) balances context richness against prompt length and processing time. Testing $k \in \{1, 2, 3, 5\}$ revealed that $k=1$ provided insufficient context with answers lacking depth, $k=2$ achieved optimal balance and was selected for final evaluation, while $k=3$ and $k=5$ showed marginal quality improvement but doubled inference time due to longer prompts.

Documents with cosine similarity below threshold τ are excluded from retrieval. Testing $\tau \in \{0.3, 0.4, 0.5, 0.6\}$ showed that $\tau=0.3$ introduced noise from marginally relevant documents,

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$\tau=0.5$ (selected value) filtered low-quality matches while retaining diverse sources, and $\tau=0.6$ was overly restrictive with some questions returning zero documents.

Retrieved chunks were truncated to prevent exceeding LLM context windows. Testing maximum characters per chunk {200, 300, 500, 1000} showed that 200 characters fragmented medical concepts with incomplete information, 300 characters (selected value) preserved key information while controlling prompt length, and 500+ characters caused timeout issues with prompts exceeding 2000 character limits.

5.2 Language Model Parameters

Temperature controls randomness in token sampling. Testing $T \in \{0.5, 0.7, 0.9\}$ showed that $T=0.5$ was overly deterministic with repetitive phrasing, $T=0.7$ (selected value) balanced consistency with natural variation, and $T=0.9$ showed excessive creativity occasionally introducing unsupported claims. Response length was limited to 400 tokens (approximately 300 words) based on medical communication guidelines recommending concise patient education materials.

5.3 Overfitting Prevention

Traditional supervised learning overfitting concerns do not directly apply to RAG systems, as there is no model training on this specific dataset. However, analogous concerns arise. To prevent retrieval overfitting where similarity threshold or top-k tuned on evaluation questions might not generalize to new queries, hyperparameters were selected using 10 held-out development questions distinct from the 25 evaluation questions, covering different clinical scenarios.

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To prevent prompt engineering overfitting where extensive prompt iteration on evaluation questions risks "teaching to the test," the system prompt was designed based on general medical communication principles rather than evaluation question performance, with only two prompt iterations conducted.

5.4 Extrapolation Detection

The 800 documents focus on refractive surgery but lack content on other ophthalmic procedures, systemic medical conditions affecting surgical outcomes, and pediatric ophthalmology. When retrieval returns no documents above threshold, the system explicitly states limitations and directs patients to consult their ophthalmologist. To prevent LLM hallucination when RAG retrieval fails, prompt instructions emphasize answering only based on provided medical references and clearly stating when information is unavailable.

6. Results

6.1 Quantitative Performance

System performance metrics show the RAG condition introduces approximately 2.4 seconds overhead, primarily from longer prompt processing rather than retrieval (which completes in 11ms). Average response time was 2.3 seconds without RAG versus 4.7 seconds with RAG. Average prompt length was 412 characters without RAG versus 1,089 characters with RAG. Response length decreased slightly with RAG (687 vs 621 characters), likely due to more focused answers grounded in specific references versus general explanations.

6.2 Physician Scoring Results

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Physician-assigned quality scores on a 5-point scale for all 25 questions showed substantial improvement with RAG. Without RAG, the mean score was 3.04 with median 3.0 and range [2, 4]. With RAG, the mean score was 4.36 with median 4.0 and range [3, 5], representing an improvement of +1.32 points or 43.4% increase (FIGURE 1).

All 25 questions showed equal or improved scores with RAG. Zero questions scored lower with RAG compared to baseline. The score distribution without RAG showed 0 questions rated excellent (5), 6 questions rated good (4), 14 questions rated adequate (3), and 5 questions rated poor (2). With RAG, the distribution improved to 12 questions rated excellent (5), 10 questions rated good (4), 3 questions rated adequate (3), and 0 questions rated poor (2). No questions in either condition were rated dangerous (1).

6.3 Qualitative Analysis

Physician feedback identified key differentiators between conditions. With RAG responses demonstrated specificity by citing exact surgical parameters (e.g., "PRK removes 50-100 microns of corneal tissue" versus vague "removes thin layer"), provided source attribution with explicit references to "Refractive Surgery PPP 2024" increasing trust and verifiability, and showed guideline alignment matching current clinical practice standards.

Without RAG responses showed weaknesses including generic language resembling patient education brochures lacking procedural specifics, temporal issues with one answer referencing a surgical technique discontinued in 2022 indicating stale training data, and overcaution with tendency toward overly general advice without preliminary guidance.

6.4 Error Analysis

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Three With RAG responses scored 3/5 instead of higher scores due to specific limitations. For the insurance coverage question, retrieved documents discussed clinical indications but lacked insurance policy information, representing a dataset gap in administrative/financial content. For the question about presbyopia interaction with prior refractive surgery, retrieved content focused on presbyopia but not its complex interaction with surgery, requiring retrieval across multiple document sections. For SMILE surgery comparison, SMILE being a newer technique had limited coverage in older PPP documents, affecting temporal knowledge base coverage.

6.5 Performance by Question Category

Performance varied by question category, showing greatest RAG benefit for safety/complications questions (+1.80 points improvement, from 2.75 to 4.55), followed by recovery timeline questions (+1.55 points, from 3.00 to 4.55), eligibility questions (+1.50 points, from 2.67 to 4.17), procedure comparison questions (+0.88 points, from 3.00 to 3.88), and procedure basics questions (+0.80 points, from 3.33 to 4.13). RAG provided greatest benefit for questions requiring specific clinical criteria and risk quantification.

6.6 Retrieval Quality Metrics

For the 25 questions, an average of 2.0 documents were retrieved per question (by design with $k=2$), with 100% retrieval success (zero questions with no retrievals). Source document distribution showed Refractive Surgery PPP 2024 dominated at 76% of retrievals, clinical studies at 16%, and surgical protocol documents at 8%, validating that the primary reference dominated retrieval as expected for a comprehensive clinical guideline.

7. Summary and Conclusions

7.1 Key Findings

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This study demonstrates that RAG substantially improves medical question-answering quality compared to standalone LLM outputs. Physician scoring showed a 43.4% improvement (3.04 → 4.36 on 5-point scale) when responses were grounded in retrieved medical documents. The system successfully retrieved relevant documents for all 25 evaluation questions, with zero cases of dangerous or misleading information. Performance gains were most pronounced for questions requiring specific clinical criteria rather than general conceptual understanding. RAG responses consistently included quantitative details and source attributions lacking in baseline outputs. The 2.4 second latency overhead from RAG is acceptable for asynchronous patient communication though may be prohibitive for real-time conversational interfaces.

7.2 Limitations

The knowledge base focuses narrowly on refractive surgery, limiting generalization to broader ophthalmology or general medicine. Questions about insurance coverage, cost, or regional practice variations fell outside dataset scope, resulting in lower-quality responses. The newest documents date from 2024, but medical knowledge evolves continuously, with newer surgical techniques receiving inadequate coverage. The 25-question evaluation set, while representative of common patient inquiries, constitutes a limited sample. Quality assessment relied on one ophthalmologist's judgment; inter-rater reliability with multiple physician reviewers would provide more robust validation. Fixed 800-character chunks occasionally split mid-sentence or separate related concepts, degrading retrieval precision. While table and image extraction were implemented, their impact was not systematically evaluated.

7.3 Lessons Learned

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The system prompt's emphasis on empathy and patient-appropriate language proved essential, as early experiments without this guidance produced technically accurate but clinically inappropriate responses. Extensive experimentation revealed 300 characters per document as optimal, as longer contexts caused LLM timeouts while shorter contexts fragmented medical concepts. The k=2 retrieval configuration struck a balance between recall and focused attention, though question complexity may warrant adaptive retrieval. DeepSeek-R1:1.5b, a 934MB model, achieved satisfactory performance for this constrained domain, suggesting RAG enables model compression for specialized applications. Source attribution significantly enhanced physician trust even when answer content was equivalent, indicating metadata should be first-class citizens in RAG design.

7.4 Future Improvements

Implementing query complexity estimation to dynamically adjust top-k would allow simple factual questions to use fewer documents while multi-faceted inquiries retrieve more context. Combining dense retrieval with keyword matching (BM25) would improve recall on exact terminology matches, as medical acronyms and drug names benefit from lexical matching. Extending the system to maintain conversation history would enable follow-up questions, as the current implementation treats each query independently. When retrieval fails or generates low-confidence responses, flagging these gaps for expert review would enable clinicians to contribute missing content, iteratively improving the knowledge base. Implementing post-hoc verification by comparing generated responses against retrieved documents using entailment models would flag statements not supported by source material. Translating the knowledge base or using multilingual LLMs would serve non-English speaking patients. Conducting user studies with actual patients would assess comprehension, trust, and behavioral impact. Combining RAG with

domain-specific fine-tuning on medical dialogue data would improve language style while using RAG for knowledge grounding.

7.5 Broader Implications

This work demonstrates RAG's potential for democratizing medical knowledge access while maintaining safety standards. By grounding LLM outputs in vetted clinical guidelines, the system reduces hallucination risks that plague general-purpose medical chatbots. The approach generalizes beyond ophthalmology to any medical specialty with well-documented practice guidelines. However, deployment in clinical settings requires addressing regulatory considerations, liability concerns, and workflow integration. The 43.4% quality improvement observed here suggests RAG can meaningfully augment, but not replace, human clinical judgment. An appropriate deployment model positions the system as a first-line resource, with human oversight for complex or ambiguous cases.

8. References

- Course Materials
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